Garden Insight

san diego, ca · spring 2026

› trust by architecture, not by policy.

your data. your agents. your control.

a lightning talk on sovereign AI infrastructure — what it is, what we're building, top research questions.

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incident · prod

the day the model broke production.

A model we did not own, on a server we did not run, changed under us — and our agents lost their minds.

[fictional illustration]

```
$ tail -f /var/log/agents/*.log

14:02:11 OK agent.researcher · tasks queued
14:18:03 OK upstream · model rev bumped
14:18:44 WARN tool calls malformed (3)
14:19:02 WARN tool calls malformed (47)
14:21:30 ERR cascading tool-call failures
14:24:00 ERR rollback unavailable

› we did not change a single line.
```

// the problem

four kinds of dependency.

01 vendor

API access can be revoked, rate-limited, or repriced. Your roadmap is downstream of theirs.

02 model

Weights change. Behavior drifts. Prompts that worked yesterday quietly degrade today.

03 data

Your inputs leave your perimeter. Compliance, IP, and customer trust become someone else’s policy.

04 cost

Per-token billing scales with usage, not value. The unit economics never quite arrive.

Key Claim

› Current agent governance approaches control the agent’s verbs.
› Regulated deployment also needs to control the data’s itinerary.

Frame sovereignty in agent systems as a triad, not a single dimension

1. sovereign data (control of flow)
2. sovereign intelligence (control of inference)
3. sovereign infrastructure (control of compute)

+1 – operator sovereignty.

Who decides governance — distinct from where bits sit, who runs the compute, or which model reasons over the data.

Key Claim

› The four dimensions are independent, non-reducible.
› Microsoft Sovereign Cloud is the closest existing analog — and still has gaps.

Why “almost” isn’t good enough

Pitch, roll, yaw, throttle — master three and you still crash.

Sovereign infrastructure	Sovereign-operator personnel, residency, disconnected operation.
Sovereign data	Customer-managed keys, in-region storage.
Sovereign intelligence	Models hosted in-region — no classification-aware routing.
Sovereign operator	Microsoft retains governance & disclosure authority over the substrate substrate

Two partials, two distinct gaps, two independent remediations. **The dimensions do not reduce.**

Source: Microsoft Sovereign Cloud documentation, accessed 2026-04-29

// policy vs architecture

the difference that matters.

under adversarial pressure, regulatory inspection, or simple human error — which one do you want to rely on?

// a policy says

"agents must not send restricted data to external model APIs."

an audit-log entry, after the fact. › detectable.

// an architecture says

"the routing layer cannot construct a request whose input contains a restricted-labeled segment."

a failure at the assembly step. › preventable.

// enforced at three points

01 // context assembly

inputs gathered under classification.

every retrieved chunk carries its label forward. assembly cannot drop or downgrade it.

02 // inference routing

architecturally constrained.

the routing layer cannot construct a request to a model whose policy disallows the input's classification.

03 // output handling

inference laundering, prevented.

outputs inherit the max of input + inferred classification. a model run on restricted data does not emit unrestricted text.

INVARIANT: inherited classification = max(input, inferred output). enforced at every boundary, not at any single boundary.

// the work

what we're building.

> we are not actually building much of this. we are composing it and contributing the missing layers + some key new thinking and capability.

an open-source reference architecture for sovereign AI — composed from open-source primitives, every layer swappable , no single vendor in the critical path.

run agents the way you run employees — *persistent identity, classified knowledge, accountable authorization, and audit trails that survive the end of any vendor relationship.*

// shape of the system

inference	local default · frontier on tap
identity	agents are principals
memory	composable by separation
governance	classification as architecture
operations	IaC-driven · rebuild target measured in minutes

// commitments

what we're committed to.

01	composability is mandatory	every layer swappable; no single vendor in the critical path.
02	agents hold authority	identity, scope, and audit live with the agent — not the prompt.
03	sovereign deployment units	a self-contained stack you can run on a rack, in a colo, or a cloud.
04	local-first, frontier on tap	route to local models by default; reach for frontier when the work demands it.

// the lab · my bench

what i have, not what it costs.

[01]
~1 TB
system ram
896 GB in service

[02]
216 GB
vram
184 GB in service

[03]
~100 TB
storage
tiered · nvme + bulk + nas

[04]
10 GiB
fabric
east-west

[disclaimer]

this is my current investment.

What it costs *your* company depends on workload, redundancy, and policy. Don't read these numbers as a requirement.

// subsystem · agentic intelligence

inference.

$ gi inspect inference

router · classify(task) → pool

local pool · on-prem

1× rtx pro 6000, 3× rtx 5090,
1× rtx 4090

primary reasoner: qwen3-32b · fp16

cloud · burstable

70b @ lower precision frontier APIs
anthropic · openai trigger: SLO / task profile

vllm · gpu-passthrough · prom metrics

solved

GPU passthrough at scale. Hybrid routing: local default, frontier on demand.

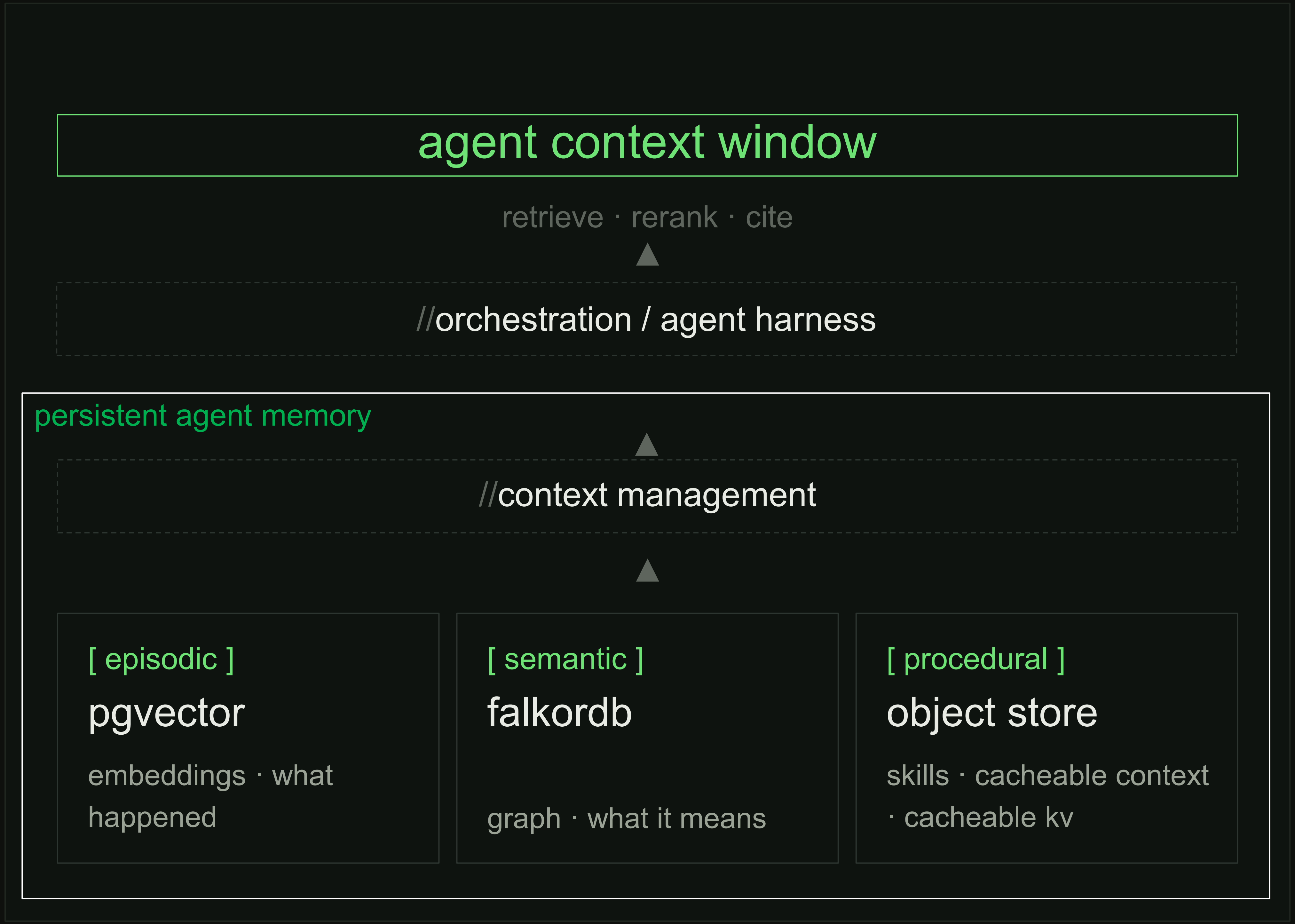
open

The local-quality gap. When does a 70b match Opus for *this* task?

● active area of research

// subsystem · context management

memory.



composable by separation

Each store owns one job. Swap any layer without rewriting the others.

solved

Storage primitives. Schema. Retrieval API. Orchestration architecture & h/l design.

open

High-speed, high-precision retrieval. Forgetting. Conflict. Hygiene.

// status

built or high-confidence design.

sensible patterns where they exist; novel work where they don't. this is a continuous research and implementation effort — not a finished product, not a final architecture.

// built / high-confidence

- [√] gpu passthrough at scale
  libvirt + vfio
- [√] hybrid inference routing
  local default, frontier on demand
- [√] governance as architecture
  classification enforced at assembly

// open · active research

- [ ] high-speed, high-precision retrieval
  graph + vector + KV at agent latency
- [ ] agent misbehavior
  detection · containment · recovery
- [ ] the local-frontier quality gap
  when does 70b suffice for this task?

[research dimension] consumer & workstation hardware, enterprise workloads — how far can prosumer iron carry real enterprise use cases?

// next

what's next.

→ k3s control plane	declarative agent fleets; rolling-update agents like pods.	
→ memory system v1	persistent agent memory deployed against the live agent population.	
→ agent scaling	· swarm fleet	5+ GPU nodes — federated with private cloud + frontier.
	· single agent	agent / operator pairing architecture(s).
→ research, build, share		

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// thank you

this is the work. happy to talk about any of it.

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questions
›_ ask anything

// reference material

appendix.

leave-behind material: subsystem inventory, model routing matrix, identity architecture, the projects we validate against.

// the lab

what's running today.

CM	context management context- and inference-optimized agent memory · falkordb + graphiti	wip		KM	knowledge management persistent organizational knowledge base	live
PM	project & program mgmt roadmaps · backlogs · kanban · incident management	live		IPA	identity · policy · audit freeipa identity live · hbac partial · dedicated audit pipeline planned	partial
SEC	secrets management openbao · approle auth · kv v2 (lab/services/infra)	live		IaC	infrastructure-as-code terraform provisions · salt configures	~90% target
ORCH	orchestration off-the-shelf today · expansion soon					minimal

5 GPU nodes
multi-host inference

3+ live agents
off-the-shelf orchestration · soon to be replaced

7 subsystems
CM · KM · PM · IPA · SEC · IaC · ORCH

// inference · policy

when each model earns its keep.

on-prem · primary · default

qwen3-32b · fp16

matches frontier on the agent-shaped work that matters.

› structured extraction · schema-bound JSON
› tool calling against stable interfaces
› RAG synthesis · in-repo code review
› agent inner loop · classify · plan · format

we aspire for 80% of the work to live here.

cloud burst · quantized

70b · fp4

cheaper to run a bigger model — until precision bites.

› good · long context, multi-step plans, broader knowledge
› risk · numeric tasks, exact recall, tight schemas
› sweet spot · reasoning where shape matters more than digits

A/B against fp16 32b before committing.

on tap · last resort

frontier · opus / gpt

large repos, complex reasoning and planning. use as needed.

› override-able by an operator when the work demands it
› otherwise: routing + cost controls decide
› every call is priced and audited

the dependency we are pricing in, not living in.

// subsystem 02 · separation of concerns

identity, secrets, lifecycle.

agents are principals, not parameters. they hold **scoped, revocable, delegable authority** under an appointed-authority model — starts human-held, transfers as trust matures. three systems own three concerns; they never overlap.

// identity & access

FreeIPA

- who/what exists (principals)
- group membership (roles)
- host access (HBAC rules)
- sudo rules per host/command

// secrets & classification

OpenBao

- dynamic, lease-based credentials
- classification enforcement
- per-task secret scoping
- org-specific tier mappings

// agent lifecycle

Orchestrator

- precision context delivery
- routing
- validator execution
- traceability enforcement
- knowledge DB hygiene

ACCESS = role ∩ host ∩ service

e.g. gi-ops ∩ hg-compute ∩ svcgrp-shell = ssh to all compute nodes

// where this fits

what we validate against.

we are not building any of these. we are composing them and contributing the missing layers between.

project	role	how we use it
FreeIPA · OpenBao	identity · secrets	the backbone of our IAM. shoulders we stand on.
FalkorDB · pgvector · postgres	graph + vector + object store	memory primitives, composed not replaced.
vLLM · llama.cpp	inference runtimes	we route to them; we do not reinvent them.
Infrastructure as Code	IaC · separation	composable tools, single-purpose, well-bounded.
k3s	control plane	where we are heading for fleet management.
Anthropic · OpenAI	frontier inference	the right call when the work demands it.